Fiscal Year:	FY 2010	**Task Last Updated:**	FY 06/15/2011
PI Name:	Hughes-Fulford, Millie Ph.D.		
Project Title:	Mechanisms and Functional Consequences of Protein Kinase C Isoform Translocation Inhibition in Monocytes Exposed to Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
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City:	San Francisco	**State:**	CA
Zip Code:	94121	**Congressional District:**	8
Comments:	PI is deceased as of February 2021		
Project Type:	Flight	**Solicitation / Funding Source:**	2001 Space Biology ILSRA
Start Date:	08/15/2002	**End Date:**	09/30/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Soyuz/ISS		
Flight Assignment:	Soyuz/ISS Sept 2006 (5/2006 update). Soyuz Taxi/ISS KUBIK Flight; Renewed. Updated Aug. 26, 2005. NOTE: Received NCE to 9/30/2010 per A. Chu/ARC (4/2009)		
Key Personnel Changes/Previous PI:	Key personnel: Tara Candelario; Chai-Fei Li ; Jim Boonratanakornkit		
COI Name (Institution):			
Grant/Contract No.:	NCC2-1361		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study investigates the mechanisms by which microgravity affects protein kinase C (PKC) signaling in monocytes and the functional consequences of these changes on downstream signaling events, gene expression and cell fate. This is important since many earthbound biological processes depend on this pathway and indeed the immune system itself is dependent on normal PKC signaling. We are analyzing the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, and are determining the functional effects of alterations in PKC signaling that occur in microgravity as well as on earth. Since all of Earths life forms have developed in a 1-g environment we will also learn if 1-g is necessary for full monocyte function. When protein kinase C isoforms are activated they translocate to the cell membrane, which is a key step in signal transduction by these kinases. Previous spaceflight experiments have shown that protein kinase C isoform translocation in monocytes & lymphocytes is inhibited in microgravity, with individual isoforms being differentially affected. The objectives of this research project are: (1) Fully characterize the effect of microgravity on gene expression, (a) The activation of key PKC induced gene expression using gene chip, (b) Verify the expression of key gene expression found in the gene chip, (c) Analyze the signaling downstream of PKC in response to phorbol ester activation, (d) Determine the effect of PKC on expression of differentiation genes in normal and microgravity. (2) Determine the effect of microgravity on PKC regulated genes which control monocyte differentiation, the initiation of apoptosis and cell cycle arrest (3) Determine if it is possible to restore agonist induced activation of individual PKC isoforms inhibited by microgravity by application of PKC isoform specific translocation enhancers and in turn determine whether these agents can restore expression of genes perturbed by microgravity. The experimental approach consists of stimulating PKC isoform activation in U937 cells using differentiation inducing agonists in microgravity and a 1.g on-board centrifuge. Gene chips will be used to identify the changes in the pattern of gene expression between microgravity, to identify which PKC isoform signaling pathways are affected by microgravity. Quantative reverse transcriptase – polymerase chain reaction (qRT-PCR) will be used to confirm expression of key genes identified from the gene array data. The sub-cellular localization of PKC isoforms and expression of key surface activation markers will be determined in formaldehyde fixed samples. Collectively, these data will allow us to explore in detail the consequences of perturbed PKC signaling in microgravity at the level of gene expression and cell fate. We will analyze the expression of patterns of gene expression induced by differentiation inducing agonists in the U937 human monocyte cell line in microgravity and 1.g, to determine the functional effects of alterations in PKC signaling that occur in microgravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Immunosuppression was seen in 15 of the 29 Apollo Astronauts during or immediately following the moon missions. Shuttle data from returning astronauts suggest that differentiation of the monocyte into the macrophage is slower in microgravity and may be one source of immunosuppression. This experiment has enabled us to identify the key genes that require gravity for gene induction as well as define key adhesion genes stimulated by the phorbol esters. This is important since many earthbound biological processes depend on this pathway and indeed the immune system itself is dependent on normal PKC signalling. We have found unique changes in microgravity on monocyte differentiation.
Task Progress:	This task is now complete; the ground based investigations are currently submitted for publication and describe changes in growth and differentiation gene expression after phorbol ester stimulation of protein kinase C on ISS in normal gravity and microgravity. We are expecting to see changes in the adhesion molecules of the monocytes in microgravity due to inhibition of differentiation in spaceflight. Final data and updates of Genespring analysis are currently being done to prepare for the manuscript submission.
Bibliography Type:	Description: (Last Updated: 03/10/2018)
Articles in Peer-reviewed Journals	Hughes-Fulford M, Li CF. "The role of FGF-2 and BMP-2 in regulation of gene induction, cell proliferation and mineralization." J Orthop Surg Res. 2011 Feb 9;6(1):8. PMID: 21306641 , Feb-2011
Articles in Peer-reviewed Journals	Hughes-Fulford M, Hatton JA, Li CF. "Gravity regulates OKC differentiation of U937 monocytes." Submitted, as of March 2011. , Mar-2011
Books/Book Chapters	Hughes-Fulford M, Boonstra JL. " Cell mechanotransduction: Cytoskeleton and related signalling pathways." in "Cell Mechanochemistry. Biological Systems and Factors Inducing Mechanical Stress, Such as Light, Pressure and Gravity." Ed. M. Monici, J. van Loon. Kerala, India : Transworld Research Network, 2010. p. 75-96., Jun-2010